Jack McMahon
B.Sc. Computer Science (COMSCI)

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Education: Third level - 2nd year results

Computer Programming 3 (Data Struct. & Alg.)	69%	Logic	73%
Computer Programming 4 (Object-Oriented & Programming)	89%	Operating Systems	58%
		Probability & Statistics	76%
Full-Stack Development	85%	Systems Analysis	64%
Introduction to Databases	68%	Systems Programming	58%
Linear Algebra	52%	Intro to DevOps	76%
Software Testing	56%	Overall Grade	70% (1.1)

Education: Second Level – Leaving Certificate

Math	H2	French	H2
English	H3	Physics	H3
Irish	H3	DCG	H1
Biology	H2		
		CAO points	543

Technical Skills

Languages:	Experienced in Java, JavaScript, Python, C, R, Haskell, Typescript
Web Technologies	HTML, CSS, JavaScript, REST APIs
Frameworks & libraries	React.js, Node.js, Django, NumPy
Dev. Ops & systems:	Git, GitHub, GitLab, Docker, Shell Scripting, CI/CD, UNIX/Linux
Databases & Modelling:	SQL, MySQL, UML, ER Diagrams, MySQL
Concepts:	Agile & Waterfall methodologies, Software Testing, Debugging

IT Projects Completed

1. **DCU Groups:** A full-stack academic group allocation platform to automate university project group formation. It has a constraint-based group allocation engine using heuristic optimization and validation logic. Integrated with Moodle LMS via REST API and developed a custom Moodle block plugin for lecturer configuration. The system was deployed to a cloud VPS (Hetzner) and configured with nginx reverse proxy and https.

2. **Custom Unix Shell:** Developed a UNIX-based shell in C supporting batch execution, I/O redirection, background processes, and command forking. Strengthened understanding of operating systems, process management, and low-level systems programming. Produced full technical documentation and user manuals.
3. **Pizza app:** Built a full-stack pizza ordering web application using Django, HTML, CSS, SQL, and Bootstrap with features such as user authentication, a customizable pizza creation page, and a checkout system. Gained hands-on experience with web frameworks and session management.
4. **Codey Viber** – AI-Assisted JavaScript Coding Web App: Developed Vibe Coder, a web application that allows users to generate, edit, and run JavaScript code through natural language prompts. The app integrates with AI APIs to interpret user instructions and dynamically produce executable code, providing an interactive coding experience.
5. **MyScrabbly:** A browser-based Scrabble clone implemented in TypeScript with a game engine, drag-and-drop tile placement, and real-time score tracking. Developed as part of a 5-person team using Agile methodology biweekly sprints and collaborative planning.

Work Experience

Odeon Cinemas

Cinema Host – Present

- Delivered exceptional customer service by handling ticket sales, welcoming guests, and upselling current offers available.
- Replenished stock during peak hours and worked together as a team efficiently to maintain fast and smooth queue flow during peak hours of the day.
- Addressed and dealt with any customer queries correctly and professionally.
- Learned key soft skills such as communication, leadership, and persuasive selling.

Fastway couriers

Courier Assistant

- Facilitated the courier with the delivery and sorting of over 100 parcels daily.
- Demonstrated strong attention to detail by verifying package information and ensuring accurate distribution.
- Provided exceptional customer support both in person and over the phone to resolve delivery issues and troubleshooting.

Interests/Volunteering/Awards/Achievements

- **PRO and Captain**, Ballyroan Abbeyleix & District AC – Demonstrated leadership and organization through coordinating various events for the club.
- **International Representation for Ireland** – Competed in the British & Irish Mountain Running Championships and the WSIT Games 2025 for the Irish team.

- **National-level Athlete** – Competed at both junior and senior national athletics championships.
- **DCU Athletics team member** – Active member of the university team, balancing academic and athletic commitments.
- **Full drivers license**.

Referees

Available on request